

Gairo Peralta

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Lead AI Engineer building tool-grounded LLM agents with evals and deterministic fallbacks, plus GeoAI on national-scale data

PROFESSIONAL EXPERIENCE

BLACKPRINT TECHNOLOGIES

Lead AI Engineer (Full-time)

Remote

2025–Present

- Led the map copilot's redesign into a tool-grounded agent (Copilot v2): **38 FastMCP** tools with per-tool authorization, **53** user-confirmed UI actions and a **778**-question blind-annotated gold eval set
- Architected a production **WhatsApp agent** that qualifies real-estate leads and books appointments via **OpenAI** tool calling, with a deterministic slot validator, human takeover and **13** golden evals
- Built **GromAI Report Studio**, grounding **Claude** site-report narratives behind a numeric firewall, claim ledger and anti-leak guard with deterministic fallback; **15** eval suites, **2,809** tests
- Built an **expansion-planning OS** demoed to **3** chains: **Claude** tool calling over deterministic services (**87%** prompt-cache hits, **58/58** Q&A audit), a finance engine within **+0.27%** of public disclosures
- Led **mireporte.ai**, which turns a map point into a 25-section market study, from **3** to **11** metro areas: per-city data contracts that fail the build on drift, sha256 manifests, **19** ADRs and **5,043** tests
- Built an **LLM-powered GeoAI platform** for **2.5M+** urban blocks and a multi-brand **site selector** (**63,724** blocks, **117** municipalities) whose v1.2 sales model cut leave-one-out MAPE from **106%** to **22%**
- Scored **acceptance propensity** of **~5.5M** merchants for a global payments network (45 GB **Postgres** to **70 MB Parquet** via **DuckDB**); reconciled **230,553 business records** with **MiniLM** embeddings
- Built logistics optimizers: an exact **MILP (HiGHS)** partner selector on a **4,851**-pair **OSRM** drive-time matrix for a **digital pharmacy**; **VRP** routing for a **pharma distributor's 3,100+** pharmacy pilot

Data Scientist

Berkeley, CA

2022–2023

- Geocoded properties via **Google Maps** (**92%** accuracy); built **pandas/SQL** ETL with **99.9%** accuracy

LAW OFFICES OF MANUEL SOLIS

Lead AI/ML Engineer (Consulting)

Remote

2025–Jul 2026

- Led an intern team building custom **LLM agents** for T-Visa declarations and cover letters, cutting turnaround from **~2 hours** to **~15 minutes** per document (**~8x faster**); automated email-to-case ingestion
- Led delivery of a **meeting-intelligence platform** for Zoom/Teams/Meet (**Speechmatics**, **GPT-5** summaries, **Pinecone** search, **3-level RBAC**) and a layout-preserving **Gemini Vision** translation pipeline

THE UNIVERSITY OF TEXAS AT SAN ANTONIO

Visiting Student Researcher (Medical AI, Volunteer)

San Antonio, TX

2025–Jan 2026

- Built a **multi-agent medical QA system** over **PrimeKG** (4M+ nodes, 33M+ edges) in **Neo4j**; multi-hop reasoning raised accuracy **+24.24 pp** to **54.69%** under systematic benchmarking

STRTGY

AI/ML Engineer (Full-time)

Remote

2023–2025

- Built a **RAG assistant** for industrial real estate (**Claude 3.5 Sonnet**, **FAISS**; **+90%** response relevance, query time down **85%**) and **GPT-4o/Vision competitor mapping** in **ArcGIS** (**+82%** decision confidence)
- Built adaptive **demand forecasting** for a beverage-industry client (**Prophet** or **SARIMA** chosen per SKU history, with inventory optimization): **+42%** forecast accuracy, **30%** lower inventory cost

PRODENSA

AI Engineer (Contract)

Remote

2024–2025

- Built **OCR + Llama 3.3 extraction** for invoices, packing lists and customs documents (**95%** accuracy, **98%** less processing time), with **FastAPI**, **React/Vite** and async PDF/CSV/XLSX ingestion

EDUCATION

TECNOLÓGICO DE MONTERREY

Master in **Applied Artificial Intelligence**

Mexico

Grade: **98/100** | **Expected 2027**

UNIVERSITY OF CALIFORNIA, BERKELEY

BA in **Data Science**; Sutardja Center for Entrepreneurship Certificate

California

2022

TECHNICAL SKILLS

LLMs & Agents: tool calling, MCP (FastMCP), RAG, evals (gold sets), grounding guards, prompt caching, LangGraph

Models & Retrieval: Claude, OpenAI (GPT-4o/GPT-5), Gemini, Llama, open-weight (Modal), Pinecone, FAISS, pgvector

GeoAI & Optimization: PostGIS, GeoPandas, H3, OSRM, isochrones, MapLibre/Mapbox GL, deck.gl, VRP, MILP (HiGHS)

ML: PyTorch, TensorFlow, scikit-learn, XGBoost, Prophet/SARIMA, LSTM/Transformers, entity resolution, LOOCV

Data/Backend: Python, TypeScript, SQL, FastAPI, Node.js, Next.js/React, Postgres, DuckDB, BigQuery, Docker, GCP